

CHASING CONSTANTS, REDUX: REFLECTION, DRIFT, AND HYBRID METRICS

ABSTRACT. An influential construction of Eberle, Guillin, and Zimmer (Transactions of the American Mathematical Society, 2019) gives quantitative contraction estimates for diffusions satisfying a Lyapunov drift condition together with a bounded-repulsion condition on the drift. The essential device is a hybrid transportation cost combining a bounded transformation of the base Euclidean metric, controlled by reflection coupling, with a Lyapunov correction that controls pairs at large separation. In this note, we isolate these two mechanisms, reduce their compatibility to two scalar inequalities, and solve the resulting optimization problem explicitly. This gives a transparent family of costs which contract with multiplicative prefactor one, together with a one-dimensional optimization over the switching radius.

1. INTRODUCTION

Quantitative convergence estimates for diffusions are particularly straightforward when the drift is globally contractive: one may couple two copies of the process synchronously and estimate their Euclidean separation directly. Many important diffusions do not satisfy such a condition. The drift may be repulsive on part of the state space, even though the process remains strongly confining at large distances and converges rapidly to equilibrium.

A standard way to treat this situation is to carefully combine these two different forms of information. A Lyapunov drift condition shows that trajectories are repeatedly drawn towards a suitable central region of the state space. A local mixing mechanism then brings together trajectories which have entered that region. This is the basic division of labour underlying Harris-type arguments.

For diffusions with non-degenerate noise, reflection coupling provides a particularly explicit local mixing mechanism. Before the two coupled trajectories meet, their Brownian increments are reflected in the direction joining them. The noise then acts directly on their separation and creates a tendency to couple. If the drift is locally repulsive, however, the basic Euclidean distance need not decrease uniformly under this coupling. One instead modifies the distance so that its concavity compensates for the possible repulsion.

The construction of Eberle, Guillin, and Zimmer combines this modified radial distance with a Lyapunov function. The radial part is responsible for contraction while the two trajectories are separated by less than a chosen switching radius. It is made bounded, and eventually constant, so that it is not required to control widely separated trajectories. Beyond the switching radius, the Lyapunov drift supplies the missing contraction. The two contributions are then joined to form a single cost on pairs of points.

There are consequently three choices to understand. One must first construct a suitable radial correction from the contraction profile of the drift. One must then choose the relative weight assigned to the radial and Lyapunov contributions. Finally, one must select the radius at which responsibility passes from the coupling estimate to the Lyapunov estimate. These choices interact, and the constants in the resulting contraction estimate depend on how they are balanced.

The purpose of this note is to make that balance as explicit as possible. We first recall the relevant assumptions and the one-dimensional differential inequality produced by reflection coupling. We then construct a bounded radial corrector for a fixed switching radius and quantify the Lyapunov decay available beyond that radius. The compatibility of the two estimates reduces to an elementary pair of scalar inequalities. Solving these inequalities yields an explicit contraction rate, after which only a one-dimensional optimization over the switching radius remains.

2. THE HYBRID CONSTRUCTION

2.1. Standing assumptions. Consider the diffusion

$$dX_t = b(X_t) dt + \sqrt{2} dB_t$$

on $\mathbf{R}^d$, with generator $L = \Delta + b \cdot \nabla$. Assume throughout that b is sufficiently regular for the diffusion and its reflection coupling to be non-explosive. Define the separation-curvature profile of b (and its repulsive part) as

$$\kappa_b(r) = \inf \left\{ -\frac{\langle b(x) - b(y), x - y \rangle}{\|x - y\|^2} : \|x - y\| = r \right\}, \quad \kappa_b^-(r) = \max \{0, -\kappa_b(r)\}.$$

We assume that $r \mapsto r \cdot \kappa_b^-(r)$ is locally integrable. Note that no assumption is made on κ_b ever being positive, and so this should be understood as a condition of ‘bounded repulsion’, rather than of contraction.

Let $V \in C^2(\mathbf{R}^d; [0, \infty))$ be coercive and suppose that for constants $C_V \geq 0$ and $\lambda_V > 0$, there holds the drift condition

$$LV \leq C_V - \lambda_V \cdot V.$$

That is, the process X_t typically drifts rather quickly into the region of the state space for which $V(x) \lesssim \frac{C_V}{\lambda_V}$.

2.2. Reflection Coupling and the Coupled Generator. Writing X_t, Y_t for our two processes to be coupled, let $\tau = \inf \{t \geq 0 : X_t = Y_t\}$ denote the coupling time. While $t < \tau$, define $E_t = \frac{X_t - Y_t}{\|X_t - Y_t\|}$ and couple the two processes according to

$$\begin{aligned} dX_t &= b(X_t) dt + \sqrt{2} dB_t, \\ dY_t &= b(Y_t) dt + \sqrt{2} \cdot (I - 2 \cdot E_t E_t^\top) dB_t. \end{aligned}$$

After time τ , the two coordinates are evolved together. Since $I - 2 \cdot E_t E_t^\top$ is an orthogonal reflection, each coordinate has the required marginal dynamics.

Write $R_t = \|X_t - Y_t\|$. Before the coupling time, the difference of the two Brownian increments is entirely directed along E_t . It follows that

$$dR_t = \langle b(X_t) - b(Y_t), E_t \rangle dt + 2\sqrt{2} d\tilde{B}_t,$$

where $\tilde{B}$ is a one-dimensional Brownian motion. By the definition of the separation-curvature profile,

$$\langle b(X_t) - b(Y_t), E_t \rangle \leq -R_t \cdot \kappa_b(R_t).$$

Thus, for a suitable nondecreasing radial function f , Itô’s formula gives the differential inequality

$$df(R_t) \leq [4 \cdot f''(R_t) - R_t \cdot \kappa_b(R_t) \cdot f'(R_t)] dt + dM_t, \quad t < \tau,$$

for a local martingale M .

Write $\tilde{L}$ for the generator of the coupled process away from the diagonal. The preceding calculation reduces the radial part of the coupling problem to the one-dimensional differential operator $f \mapsto 4 \cdot f'' - r \cdot \kappa_b \cdot f'$. The following lemma records this estimate together with the corresponding estimate for the two-particle Lyapunov function.

Throughout, to say that a cost contracts exactly at rate c means that its contraction estimate holds with multiplicative prefactor one; it does not assert that c is an optimal rate.

Lemma 1. *Let (X_t, Y_t) be the reflection coupling, made absorbing after its coupling time, and put $R_t = \|X_t - Y_t\|$. If f is nonnegative, nondecreasing and concave, with $f(0) = 0$, and if*

$$\tilde{f}(x, y) = f(\|x - y\|),$$

then, before the coupling time,

$$(2.1) \quad (\tilde{L}\tilde{f})(x, y) \leq 4 \cdot f''(r) - r \cdot \kappa_b(r) \cdot f'(r), \quad r = \|x - y\|.$$

Moreover, defining $\tilde{V}(x, y) = V(x) + V(y)$, it holds that

$$(2.2) \quad (\tilde{L}\tilde{V})(x, y) = LV(x) + LV(y) \leq 2 \cdot C_V - \lambda_V \cdot \tilde{V}(x, y).$$

Proof. Before the coupling time, reflection coupling gives

$$dR_t = \left\langle b(X_t) - b(Y_t), \frac{X_t - Y_t}{R_t} \right\rangle dt + 2\sqrt{2} d\beta_t.$$

By the definition of κ_b , $\left\langle b(X_t) - b(Y_t), \frac{X_t - Y_t}{R_t} \right\rangle \leq -\kappa_b(R_t) \cdot R_t$, and application of Itô’s formula therefore gives (2.1).

The estimate (2.2) follows because each component of the coupling has generator L , and the function $\tilde{V}$ contains no mixed derivatives. $\square$

2.3. The Bounded Radial Corrector. Fix a switching radius $R > 0$. We seek a nondecreasing concave function f for which

$$4 \cdot f''(r) - r \cdot \kappa_b(r) \cdot f'(r) \leq -a$$

on $0 < r < R$, for some explicit constant $a > 0$. We also require $f'(0) = 1$ (as a normalisation) and $f'(R) = 0$ (so that f can be continued as a constant on $[R, \infty)$). Writing $g = f'$, observe that the positive part of κ_b is favourable from a contraction perspective. It is therefore natural to construct g by enforcing equality only against the repulsive part of the curvature profile:

$$4 \cdot g'(r) + r \cdot \kappa_b^-(r) \cdot g(r) = -a, \quad g(0) = 1.$$

Indeed, whenever $g \geq 0$,

$$\begin{aligned} 4 \cdot g' - r \cdot \kappa_b \cdot g &= -a - r \cdot (\kappa_b^- + \kappa_b) \cdot g \\ &= -a - r \cdot \kappa_b^+ \cdot g \\ &\leq -a. \end{aligned}$$

The integrating factor for this first-order equation is $\exp(J_b(r))$, where $J_b(r) = \frac{1}{4} \cdot \int_0^r s \cdot \kappa_b^-(s) \, ds$, and we may thus take

$$g(r) = \exp(-J_b(r)) \left[1 - \frac{a}{4} \cdot I_b(r) \right], \quad I_b(r) = \int_0^r \exp(J_b(s)) \, ds.$$

The boundary condition $g(R) = 0$ determines the constant as $a = \frac{4}{I_b(R)}$. The corrector appearing below is obtained by integrating this g and extending it with zero derivative beyond R . In this sense, once the switching radius has been chosen, the formula is the direct solution of a first-order boundary-value problem rather than an additional ansatz.

Lemma 2. For $r \geq 0$, define

$$J_b(r) = \frac{1}{4} \cdot \int_0^r s \cdot \kappa_b^-(s) \, ds, \quad I_b(r) = \int_0^r \exp(J_b(s)) \, ds.$$

Fix $R > 0$, and set

$$g_{b,R}(r) = \begin{cases} \exp(-J_b(r)) \cdot \left(1 - \frac{I_b(r)}{I_b(R)} \right), & 0 \leq r \leq R, \\ 0, & r \geq R. \end{cases}$$

Define

$$f_{b,R}(r) = \int_0^r g_{b,R}(s) \, ds, \quad F_b(R) = f_{b,R}(R), \quad a_b(R) = \frac{4}{I_b(R)}.$$

Then $f_{b,R}$ is nonnegative, nondecreasing, concave and constant on $[R, \infty)$. Furthermore,

$$(2.3) \quad 4 \cdot f_{b,R}''(r) - r \cdot \kappa_b(r) \cdot f_{b,R}'(r) \leq -a_b(R), \quad 0 < r < R.$$

Consequently, for

$$\bar{f}_{b,R} = \frac{f_{b,R}}{F_b(R)}, \quad \gamma_b(R) = \frac{a_b(R)}{F_b(R)},$$

one has $0 \leq \bar{f}_{b,R} \leq 1$ and

$$(2.4) \quad 4 \cdot \bar{f}_{b,R}''(r) - r \cdot \kappa_b(r) \cdot \bar{f}_{b,R}'(r) \leq -\gamma_b(R), \quad 0 < r < R.$$

Proof. Differentiation shows that, for $0 < r < R$,

$$g_{b,R}'(r) = -\frac{1}{4} \cdot r \cdot \kappa_b^-(r) \cdot g_{b,R}(r) - \frac{1}{I_b(R)}.$$

In particular, $g_{b,R} \geq 0$, $g_{b,R}(R) = 0$, and $g_{b,R}' < 0$. Hence $f_{b,R}$ is increasing and concave.

Moreover, compute that

$$\begin{aligned} 4 \cdot g_{b,R}'(r) - r \cdot \kappa_b(r) \cdot g_{b,R}(r) &= -a_b(R) - r \cdot (\kappa_b(r) + \kappa_b^-(r)) \cdot g_{b,R}(r) \\ &= -a_b(R) - r \cdot (\kappa_b(r))_+ \cdot g_{b,R}(r) \\ &\leq -a_b(R), \end{aligned}$$

which proves (2.3); the estimate (2.4) then follows by normalization. $\square$

Lemma 3. *For every $R > 0$,*

$$\gamma_b(R) \geq \frac{4}{R^2} \cdot \exp\left(-\frac{1}{4} \cdot \int_0^R s \cdot \kappa_b^-(s) \, ds\right).$$

If $\kappa_b \geq 0$ on $[0, R]$, then $g_{b,R}(r) = 1 - \frac{r}{R}$, $F_b(R) = \frac{R}{2}$, $a_b(R) = \frac{4}{R}$, and therefore $\gamma_b(R) = \frac{8}{R^2}$.

Proof. Since J_b is nondecreasing, $I_b(R) \leq R \cdot \exp(J_b(R))$, and hence $a_b(R) \geq \frac{4}{R} \cdot \exp(-J_b(R))$. Observing that $0 \leq g_{b,R} \leq 1$, it follows that $F_b(R) \leq R$. If $\kappa_b^- = 0$ on $[0, R]$, then $J_b = 0$ and $I_b(r) = r$, from which the remaining claims follow by direct calculation. $\square$

2.4. Lyapunov Control at Large Separation. The radial corrector is designed to be constant once the separation reaches R . This is intentional, and reflects the intuition that while the reflection coupling has the chance to induce good contraction on small spatial scales, on larger scales, one should instead take advantage of different features of the dynamics. We thus identify a second mechanism to obtain strict contraction in that region.

The Lyapunov drift condition provides this mechanism. Under the coupled dynamics,

$$\tilde{L}[V(x) + V(y)] \leq 2 \cdot C_V - \lambda_V \cdot (V(x) + V(y)).$$

The constant term prevents this from being a homogeneous contraction estimate everywhere. For widely separated pairs, however, coercivity ensures that at least one of $V(x)$ and $V(y)$ is large. The negative term can then absorb the constant defect $2 \cdot C_V$.

The next lemmas quantify this observation. They measure the smallest value of $V(x) + V(y)$ among pairs whose separation is at least R , and use this value to convert the Lyapunov inequality into a homogeneous decay estimate on the outer region. This allows us to identify the point at which ‘responsibility’ for the contraction ought to pass from the radial corrector to the Lyapunov drift.

Lemma 4. *Define*

$$m_V(r) = \inf \{V(x) + V(y) : \|x - y\| = r\},$$

$$M_V(R) = \inf \{m_V(r) : r \geq R\}.$$

Then M_V is nondecreasing and $M_V(R) \rightarrow \infty$ as $R \rightarrow \infty$.

Whenever $M_V(R) > 0$, define $\delta_V(R) = \frac{2 \cdot C_V}{M_V(R)}$. For every x, y satisfying $\|x - y\| \geq R$, it holds that

$$(2.5) \quad 2 \cdot C_V - \lambda_V \cdot (V(x) + V(y)) \leq -(\lambda_V - \delta_V(R)) \cdot (V(x) + V(y)).$$

Proof. Monotonicity of M_V is immediate. If $\|x - y\| \geq R$, then $\max\{\|x\|, \|y\|\} \geq \frac{R}{2}$. Since $V \geq 0$ is coercive, $M_V(R) \geq \inf \{V(z) : \|z\| \geq \frac{1}{2} \cdot R\}$ diverges as R grows. Finally, if $\tilde{V}(x, y) = V(x) + V(y)$, then $\tilde{V} \geq M_V(R)$, and hence

$$2 \cdot C_V = \delta_V(R) \cdot M_V(R) \leq \delta_V(R) \cdot \tilde{V}.$$

Substitution gives (2.5). $\square$

2.5. The Hybrid Cost and its two regimes. We now combine the radial and Lyapunov corrections. For a weight $\eta > 0$, consider

$$\rho_{b,V}^{R,\eta}(x, y) = [\bar{f}_{b,R}(\|x - y\|) + \eta \cdot (V(x) + V(y))] \cdot \mathbf{1}_{\{x \neq y\}}.$$

The indicator makes the cost vanish on the diagonal. Although the Lyapunov contribution does not tend to zero as y approaches x , this causes no difficulty for the coupling argument: when the two coordinates meet, the cost jumps downwards to zero and remains there.

The conditions needed for contraction can be read off separately in the inner and outer regions. Write $\tilde{V} : (x, y) \mapsto V(x) + V(y)$, and fix a proposed rate $c \in (0, \lambda_V)$. When $0 < \|x - y\| < R$, the radial corrector gives

$$\tilde{L}\bar{f}_{b,R}(\|x - y\|) \leq -\gamma_b(R).$$

Using $0 \leq \bar{f}_{b,R} \leq 1$, $\tilde{V} \geq 0$, and $c < \lambda_V$, one obtains

$$\tilde{L}\rho_{b,V}^{R,\eta} + c \cdot \rho_{b,V}^{R,\eta} \leq -\gamma_b(R) + 2 \cdot \eta \cdot C_V + c.$$

It is therefore sufficient that $c + 2 \cdot \eta \cdot C_V \leq \gamma_b(R)$.

When $\|x - y\| \geq R$, the normalized radial corrector is equal to one and has zero generator. Since $\tilde{V} \geq M_V(R)$,

$$\tilde{L}\rho_{b,V}^{R,\eta} + c \cdot \rho_{b,V}^{R,\eta} \leq c + 2 \cdot \eta \cdot C_V - \eta \cdot (\lambda_V - c) \cdot M_V(R).$$

It is therefore sufficient that

$$c + 2 \cdot \eta \cdot C_V \leq \eta \cdot (\lambda_V - c) \cdot M_V(R).$$

The first condition prevents the Lyapunov defect from exhausting the contraction supplied by reflection coupling. The second ensures that the Lyapunov term is sufficiently heavily weighted to control the outer region. The following proposition records this inner–outer argument.

Proposition 5. *Fix $R > 0$, $\eta > 0$, and $c \in (0, \lambda_V)$. Define*

$$\rho_{b,V}^{R,\eta}(x, y) = [\bar{f}_{b,R}(\|x - y\|) + \eta \cdot (V(x) + V(y))] \mathbf{1}_{\{x \neq y\}}.$$

Suppose that

$$(2.6) \quad \begin{aligned} c + 2 \cdot \eta \cdot C_V &\leq \gamma_b(R), \\ c + 2 \cdot \eta \cdot C_V &\leq \eta \cdot (\lambda_V - c) \cdot M_V(R). \end{aligned}$$

Then, away from the diagonal, it holds that

$$\tilde{L}\rho_{b,V}^{R,\eta} \leq -c \cdot \rho_{b,V}^{R,\eta}.$$

Proof. Write $\tilde{V}(x, y) = V(x) + V(y)$, $r = \|x - y\|$. By Lemma 1,

$$\begin{aligned} \tilde{L}\rho_{b,V}^{R,\eta} + c \cdot \rho_{b,V}^{R,\eta} &\leq 4 \cdot \bar{f}_{b,R}''(r) - r \cdot \kappa_b(r) \cdot \bar{f}_{b,R}'(r) + c \cdot \bar{f}_{b,R}(r) \\ &\quad + 2 \cdot \eta \cdot C_V - \eta \cdot (\lambda_V - c) \cdot \tilde{V}(x, y). \end{aligned}$$

If $0 < r < R$, Lemma 2, $0 \leq \bar{f}_{b,R} \leq 1$, and $\tilde{V} \geq 0$ give

$$\tilde{L}\rho_{b,V}^{R,\eta} + c \cdot \rho_{b,V}^{R,\eta} \leq -\gamma_b(R) + c + 2 \cdot \eta \cdot C_V \leq 0.$$

If $r \geq R$, then $\bar{f}_{b,R}(r) = 1$, its generator contribution vanishes, and $\tilde{V}(x, y) \geq M_V(R)$. Consequently,

$$\tilde{L}\rho_{b,V}^{R,\eta} + c \cdot \rho_{b,V}^{R,\eta} \leq c + 2 \cdot \eta \cdot C_V - \eta \cdot (\lambda_V - c) \cdot M_V(R) \leq 0,$$

and the claim follows. $\square$

2.6. Balancing the Constants. For a fixed switching radius, the probabilistic part of the argument is now complete. It remains to determine whether the radial and Lyapunov estimates can be made compatible by a suitable choice of the weight η .

To isolate this algebraic problem, write $\gamma = \gamma_b(R)$, $\lambda = \lambda_V$, $M = M_V(R)$, $C = C_V$, and $\delta = \frac{2 \cdot C}{M}$. The weight η is constrained in opposite directions. The inner-region condition gives the upper bound $\eta \leq \frac{\gamma - c}{2 \cdot C}$. By contrast, the outer-region condition can be written as $c \leq \eta \cdot M \cdot (\lambda - c - \delta)$, and hence gives the lower bound $\eta \geq \frac{c}{M \cdot (\lambda - c - \delta)}$. A large weight helps thus in the outer region, but increases the Lyapunov defect which must be absorbed in the inner region.

The condition $\delta < \lambda$ ensures that the Lyapunov estimate retains a strictly positive effective decay rate sufficiently far from the diagonal. For a proposed rate c , the upper and lower bounds on η are compatible precisely when

$$\frac{c}{M \cdot (\lambda - c - \delta)} \leq \frac{\gamma - c}{2 \cdot C}.$$

Using $2 \cdot C = M \cdot \delta$, this is equivalent to

$$(\gamma - c) \cdot (\lambda - c) \geq \gamma \cdot \delta.$$

The largest feasible rate is obtained when equality holds. The square-root expression in the following lemma is therefore simply the smaller root of this quadratic equation. At that rate, the upper and lower restrictions on η coincide.

Lemma 6. *Let $\gamma > 0$, $\lambda > 0$, $M > 0$, $C > 0$, and $\delta = \frac{2 \cdot C}{M} < \lambda$. Consider the inequalities*

$$(2.7) \quad \begin{aligned} c + 2 \cdot \eta \cdot C &\leq \gamma, \\ c + 2 \cdot \eta \cdot C &\leq \eta \cdot (\lambda - c) \cdot M. \end{aligned}$$

There exists $\eta > 0$ satisfying (2.7) if and only if

$$0 < c \leq c_*(\gamma, \lambda, \delta) := \frac{\gamma + \lambda - \sqrt{(\gamma - \lambda)^2 + 4 \cdot \gamma \cdot \delta}}{2}.$$

At $c = c_$, the unique balancing weight is*

$$\eta_* = \frac{c_*}{M \cdot (\lambda - c_* - \delta)} = \frac{\gamma - c_*}{2 \cdot C}.$$

Proof. The inequalities in (2.7) are equivalent to

$$\frac{c}{M \cdot (\lambda - c - \delta)} \leq \eta \leq \frac{\gamma - c}{2 \cdot C}.$$

The interval is nonempty precisely when $(\gamma - c) \cdot (\lambda - c) \geq \gamma \cdot \delta$. The smaller root of the corresponding quadratic equation is the given algebraic expression. Since $\delta < \lambda$, it lies in $(0, \min\{\gamma, \lambda - \delta\})$, and the feasible rates form the interval $(0, c_\star]$. At the endpoint both inequalities in the system hold with equality, giving the claim. $\square$

Theorem 7. Fix $R > 0$ such that $M_V(R) > 0$ and $\delta_V(R) < \lambda_V$. Define

$$\begin{aligned} c_{b,V}(R) &= \frac{\gamma_b(R) + \lambda_V - \sqrt{(\gamma_b(R) - \lambda_V)^2 + 4 \cdot \gamma_b(R) \cdot \delta_V(R)}}{2} \\ \eta_{b,V}(R) &= \frac{c_{b,V}(R)}{M_V(R) \cdot (\lambda_V - c_{b,V}(R) - \delta_V(R))} \\ \rho_{b,V}^R(x, y) &= [\bar{f}_{b,R}(\|x - y\|) + \eta_{b,V}(R) \cdot (V(x) + V(y))] \mathbf{1}_{\{x \neq y\}}. \end{aligned}$$

Then $\rho_{b,V}^R$ contracts exactly at rate $c_{b,V}(R)$:

$$\mathbf{E} [\rho_{b,V}^R(X_t, Y_t)] \leq \exp(-c_{b,V}(R) \cdot t) \cdot \rho_{b,V}^R(x, y).$$

Consequently, if

$$\mathcal{T}_\rho(\mu, \nu) = \inf_{\pi \in \Pi(\mu, \nu)} \int \rho(x, y) \pi(dx, dy),$$

then, for probability measures with finite V -moment,

$$\mathcal{T}_{\rho_{b,V}^R}(\mu P_t, \nu P_t) \leq \exp(-c_{b,V}(R) \cdot t) \cdot \mathcal{T}_{\rho_{b,V}^R}(\mu, \nu).$$

Proof. Lemma 6 shows that the choices described satisfy (2.6). Proposition 5 therefore gives

$$\tilde{L}\rho_{b,V}^R \leq -c_{b,V}(R) \cdot \rho_{b,V}^R$$

before the coupling time. Apply Itô's formula to

$$\exp(c_{b,V}(R) \cdot t) \cdot \rho_{b,V}^R(X_t, Y_t)$$

up to the coupling time and a standard localization sequence. At the coupling time the cost drops to zero, which contributes with the favourable sign. The Lyapunov estimate supplies the required integrability for removing the localization, and the first result follows. Starting from an arbitrary coupling of μ and ν , run the reflection coupling conditionally on its initial pair. The resulting joint law at time t is a coupling of μP_t and νP_t , and so the transportation-cost estimate follows by integration. $\square$

Corollary 8. Define

$$c_{b,V}^\star = \sup \{c_{b,V}(R) : M_V(R) > 0, \delta_V(R) < \lambda_V\}.$$

Then every $0 < c < c_{b,V}^\star$ is an exact contraction rate of a cost of the form

$$[\bar{f}_{b,R}(\|x - y\|) + \eta \cdot (V(x) + V(y))] \mathbf{1}_{\{x \neq y\}}$$

for a suitable choice of R and η .

If the supremum defining $c_{b,V}^\star$ is attained at $R_{b,V}^\star$, then $c_{b,V}^\star$ itself is an exact contraction rate.

Proof. For every $0 < c < c_{b,V}^\star$, there exists R such that $c < c_{b,V}(R)$. Lemma 6 supplies a finite admissible η , and Theorem 7 gives the claimed contraction. $\square$

Proposition 9. For $\gamma > 0$, $\lambda > 0$, and $0 < \delta < \lambda$, let

$$\mathfrak{c}(\gamma, \lambda, \delta) = \frac{\gamma + \lambda - \sqrt{(\gamma - \lambda)^2 + 4 \cdot \gamma \cdot \delta}}{2}.$$

There holds the bound

$$0 < \mathfrak{c}(\gamma, \lambda, \delta) < \min\{\gamma, \lambda - \delta\}.$$

On this region, $\mathfrak{c}$ is strictly increasing in γ and λ , and strictly decreasing in δ . On the full region $0 < \delta < \lambda$, it is nondecreasing in γ and λ , and strictly decreasing in δ . Moreover, the following limiting identities hold:

$$\begin{aligned}\lim_{\delta \rightarrow 0} \mathfrak{c}(\gamma, \lambda, \delta) &= \min\{\gamma, \lambda\}, \\ \lim_{\gamma \rightarrow \infty} \mathfrak{c}(\gamma, \lambda, \delta) &= \lambda - \delta, \\ \lim_{\lambda \rightarrow \infty} \mathfrak{c}(\gamma, \lambda, \delta) &= \gamma.\end{aligned}$$

In the balanced case $\gamma = \lambda$,

$$\mathfrak{c}(\lambda, \lambda, \delta) = \lambda - \sqrt{\lambda \cdot \delta}.$$

Proof. The rate $c = \mathfrak{c}(\gamma, \lambda, \delta)$ is characterized as the smaller solution of

$$(\gamma - c) \cdot (\lambda - c) = \gamma \cdot \delta.$$

The bounds and limiting identities follow directly from this equation. When $\delta > 0$, writing $D = \gamma + \lambda - 2 \cdot c > 0$, implicit differentiation gives

$$\frac{\partial c}{\partial \gamma} = \frac{\lambda - c - \delta}{D} > 0, \quad \frac{\partial c}{\partial \lambda} = \frac{\gamma - c}{D} > 0, \quad \frac{\partial c}{\partial \delta} = -\frac{\gamma}{D} < 0,$$

from which the monotonicity claims follow. $\square$

3. SOME EXAMPLES

Example 10. Suppose that there exist constants $A \geq 0$, $B > 0$, and $L_b \geq 0$ such that globally, $\langle b(x), x \rangle \leq A - B \cdot \|x\|^2$ and $\kappa_b(r) \geq -L_b$. Take $V(x) = \|x\|^2$. Since $L = \Delta + b \cdot \nabla$, compute that

$$LV(x) = 2 \cdot d + 2 \cdot \langle b(x), x \rangle \leq 2 \cdot (d + A) - 2 \cdot B \cdot V(x),$$

whereby one may take $C_V = 2 \cdot (d + A)$, $\lambda_V = 2 \cdot B$.

The parallelogram identity gives that $\|x\|^2 + \|y\|^2 = \frac{1}{2} \cdot \|x - y\|^2 + \frac{1}{2} \cdot \|x + y\|^2$, and so one can take $M_V(R) = \frac{R^2}{2}$, $\delta_V(R) = \frac{8 \cdot (d + A)}{R^2}$; the condition $\delta_V(R) < \lambda_V$ is therefore equivalent to $R^2 > \frac{4 \cdot (d + A)}{B}$.

The curvature assumption implies that $J_b(R) \leq \frac{L_b \cdot R^2}{8}$. By Lemma 3, deduce that $\gamma_b(R) \geq \frac{4}{R^2} \cdot \exp\left(-\frac{L_b \cdot R^2}{8}\right)$. Since the rate function $\mathfrak{c}$ is nondecreasing in its first argument, it follows that

$$c_{b,V}(R) \geq \mathfrak{c}\left(\frac{4}{R^2} \cdot \exp\left(-\frac{L_b \cdot R^2}{8}\right), 2 \cdot B, \frac{8 \cdot (d + A)}{R^2}\right).$$

Hence, for every R satisfying $R^2 > \frac{4 \cdot (d + A)}{B}$, every positive rate strictly below the right-hand side is an exact contraction rate for a corresponding hybrid cost.

Example 11. Consider in dimension one the diffusion with drift $b(x) = a \cdot x - x^3$, with $a > 0$. For $x \neq y$,

$$-\frac{b(x) - b(y)}{x - y} = x^2 + x \cdot y + y^2 - a \geq \frac{1}{4} \cdot |x - y|^2 - a,$$

i.e. one can take $\kappa_b(r) = \frac{r^2}{4} - a$, $\kappa_b^-(r) = \left(a - \frac{r^2}{4}\right) \cdot \mathbf{1}_{\{r < 2\sqrt{a}\}}$. It follows that

$$J_b(r) = \begin{cases} \frac{1}{8} \cdot a \cdot r^2 - \frac{1}{64} \cdot r^4, & 0 \leq r \leq 2\sqrt{a}, \\ \frac{1}{4} \cdot a^2, & r \geq 2\sqrt{a}. \end{cases}$$

Take $V(x) = x^2$. For $\lambda > 0$, bound

$$LV(x) = 2 + 2 \cdot a \cdot x^2 - 2 \cdot x^4 \leq C_V(\lambda) - \lambda \cdot V(x)$$

with $C_V(\lambda) = 2 + \frac{(2 \cdot a + \lambda)^2}{8}$. As before, $M_V(R) = \frac{R^2}{2}$, and hence

$$\delta_V(R) = \frac{4 \cdot C_V(\lambda)}{R^2} = \frac{1}{R^2} \cdot \left[8 + \frac{(2 \cdot a + \lambda)^2}{2}\right].$$

For $R \geq 2\sqrt{a}$, the radial estimate thus becomes $\gamma_b(R) \geq \frac{4}{R^2} \cdot \exp\left(-\frac{1}{4} \cdot a^2\right)$. Taking now $E_a = \exp\left(-\frac{1}{4} \cdot a^2\right)$, $G_a = 4 \cdot E_a$, $D_{a,\lambda} = 8 + \frac{(2 \cdot a + \lambda)^2}{2}$ and writing $q = R^{-2}$, we obtain the explicit lower bound

$$c_{b,V}(R) \geq \mathfrak{c}(G_a \cdot q, \lambda, D_{a,\lambda} \cdot q).$$

For fixed λ , this lower bound is maximized at $q_\star = \frac{2 \cdot \lambda}{G_a + 4 \cdot D_{a,\lambda}}$, or equivalently, where $R_\star^2 = \frac{2 \cdot E_a + 16 + (2 \cdot a + \lambda)^2}{\lambda}$. This value automatically satisfies $R_\star \geq 2\sqrt{a}$. The resulting optimized lower bound is then

$$\underline{c}_a(\lambda) = \frac{\lambda \cdot E_a}{E_a + 8 + \frac{(2 \cdot a + \lambda)^2}{2}}.$$

The remaining optimization over the Lyapunov rate is also explicit. Differentiation shows that $\underline{c}_a(\lambda)$ is maximized at $\lambda_\star = \sqrt{4 \cdot a^2 + 16 + 2 \cdot E_a}$. At this choice, $R_\star^2 = 2 \cdot (\lambda_\star + 2 \cdot a)$, and

$$\underline{c}_a^\star = \frac{\exp(-\frac{1}{4} \cdot a^2)}{2 \cdot a + \sqrt{4 \cdot a^2 + 16 + 2 \cdot \exp(-\frac{1}{4} \cdot a^2)}},$$

whereby every $0 < c < \underline{c}_a^\star$ is an exact contraction rate for a corresponding hybrid cost.

Example 12. Consider the drift $b(x) = -m \cdot x + u(x)$, $m > 0$ where $\|u(x)\| \leq K$ and $\langle u(x) - u(y), x - y \rangle \leq L \cdot \|x - y\|^2$, so that $\kappa_b(r) \geq m - L$. Defining $q_b = \max\{0, L - m\}$, one therefore has $\kappa_b^-(r) \leq q_b$, $J_b(R) \leq \frac{q_b \cdot R^2}{8}$.

Take then $V(x) = \|x\|^2$. Young's inequality gives $2 \cdot \langle x, u(x) \rangle \leq m \cdot \|x\|^2 + \frac{K^2}{m}$, and consequently

$$LV(x) = 2 \cdot d - 2 \cdot m \cdot \|x\|^2 + 2 \cdot \langle x, u(x) \rangle \leq \left(2 \cdot d + \frac{K^2}{m}\right) - m \cdot V(x),$$

i.e. one may take $C_V = 2 \cdot d + \frac{K^2}{m}$, $\lambda_V = m$.

Since $M_V(R) = \frac{R^2}{2}$, one has $\delta_V(R) = \frac{4 \cdot C_V}{R^2}$. Choose $R^2 = \frac{8 \cdot C_V}{m}$, so that $\delta_V(R) = \frac{m}{2}$, while Lemma 3 gives

$$\gamma_b(R) \geq \frac{4}{R^2} \cdot \exp\left(-\frac{q_b \cdot R^2}{8}\right) = \frac{m}{2 \cdot C_V} \cdot \exp\left(-\frac{q_b \cdot C_V}{m}\right) =: \gamma_0.$$

By monotonicity of the rate function in its first argument, see that

$$\begin{aligned} c_{b,V}(R) &\geq \mathfrak{c}\left(\gamma_0, m, \frac{m}{2}\right) \\ &= \frac{\gamma_0 + m - \sqrt{\gamma_0^2 + m^2}}{2} \\ &= \frac{\gamma_0 \cdot m}{\gamma_0 + m + \sqrt{\gamma_0^2 + m^2}}. \end{aligned}$$

Since $C_V \geq 2$, one has $\gamma_0 \leq \frac{m}{2 \cdot C_V} \leq \frac{m}{4}$. It follows that

$$\gamma_0 + m + \sqrt{\gamma_0^2 + m^2} \leq \frac{5 + \sqrt{17}}{4} \cdot m,$$

and hence $c_{b,V}(R) \geq \frac{4}{5 + \sqrt{17}} \cdot \gamma_0$. Substituting the definition of γ_0 gives the completely explicit estimate

$$c_{b,V}(R) \geq \frac{2 \cdot m}{(5 + \sqrt{17}) \cdot (2 \cdot d + K^2/m)} \cdot \exp\left(-\frac{\max\{0, L - m\}}{m} \left(2 \cdot d + \frac{K^2}{m}\right)\right).$$

4. CONCLUSION

The construction separates the contraction argument into two mechanisms. Reflection coupling supplies a bounded radial corrector with strength $\gamma_b(R)$, appropriate for displaying contraction over small spatial scales, while the Lyapunov condition supplies an effective outer decay rate determined by λ_V and $\delta_V(R)$, accurately characterising the favourable large-scale equilibration of the process. The weight η balances these mechanisms, and the square-root formula for the contraction rate results from making the corresponding inner and outer constraints simultaneously active.

The switching radius R is the genuine free parameter of the construction. A small value of R favours the radial corrector but may leave too large a Lyapunov defect. A large value improves the effective Lyapunov estimate but generally weakens the local coupling constant. The optimization over R is therefore a transparent one-dimensional version of the constant-selection problem in the original argument.

The rate formula also makes clear how the estimate may be improved. A sharper radial construction increases $\gamma_b(R)$; a stronger Lyapunov drift increases λ_V ; and a better lower bound on $M_V(R)$ decreases $\delta_V(R)$. The monotonicity properties of $\mathfrak{c}$ then translate each of these improvements directly into a better contraction rate. This modularity is one of the main advantages of isolating the algebraic balance from the probabilistic estimates.

Finally, suppose that π is an invariant probability measure with finite V -moment. For any cost ρ constructed above which contracts exactly at rate c , one immediately obtains

$$\mathcal{T}_\rho(\mu P_t, \pi) \leq \exp(-c \cdot t) \cdot \mathcal{T}_\rho(\mu, \pi).$$

The same estimate shows uniqueness of the invariant measure within the class of probability measures having finite V -moment. Further comparisons with weighted total-variation distances, gradient estimates, or more standard Wasserstein distances require additional hypotheses. The purpose of the present note has instead been to isolate, simplify, and optimize the core coupling construction from which such consequences follow.